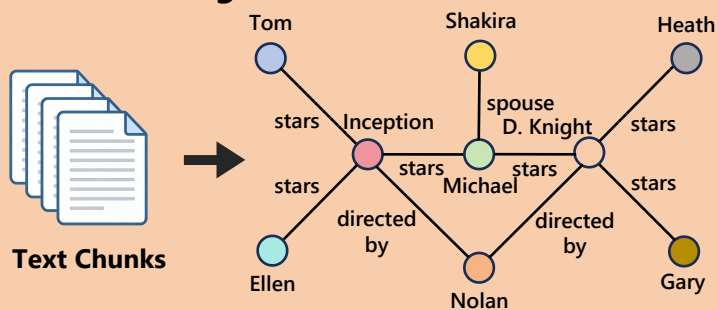


Fine-grained Fact Extraction



Graph-grounded Questions Construction

[?] - stars - Michael (mask Inception)

What movie does Michael Caine stars in?

Inception - stars - [?] (mask Michael)

Who stars Inception?

[?] - stars - Michael (mask D. Knight)

What movie does Michael Caine stars in?

D. Knight - stars - [?] (mask Michael)

Who stars The Dark Knight?

Inception - stars - [?] (mask Tom)

Who stars Inception?

Michael - stars - [?] (mask Shakira)

Who is Michael Caine spouse?

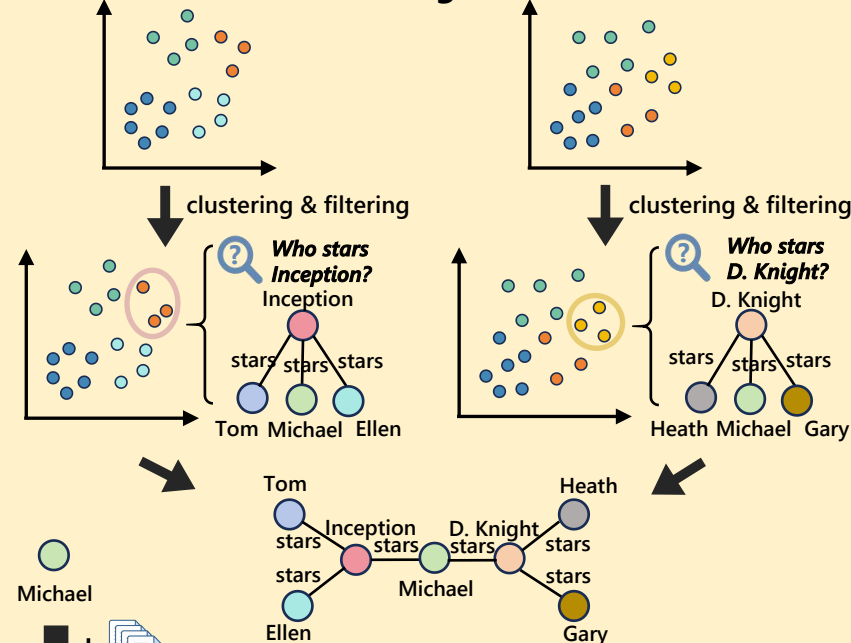
....

User Query: Who is the spouse of the actor who appeared in both Inception and The Dark Knight?

Atomic Sub-query 1: Who stars Inception?

Atomic Sub-query 2: Who stars The Dark Knight?

Holistic Knowledge Retrieval



Sufficiency Triggered Inference

Atomic Sub-query 3: Who is the spouse of Michael Caine?

Dense Retrieval



insufficient to answer user query

Query Rewrite

Tok-K Questions



Answer: Shakira



sufficient to answer user query

